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MAST20030: Differential Equations
Exam Solutions S2 2022

Question 1.

Solution.

(a) (i) Substituting x^λ into the ODE gives

working+1
solution+1

$$\begin{aligned} x^2\lambda(\lambda-1)x^{\lambda-2} + x\lambda x^{\lambda-1} - n^2x^\lambda &= 0 \\ \lambda(\lambda-1)x^\lambda + \lambda x^\lambda - n^2x^\lambda &= 0 \\ \lambda(\lambda-1) + \lambda - n^2 &= 0 \\ \lambda^2 &= n^2 \end{aligned}$$

and so $\lambda = \pm n$ which gives the general solution $y(x) = Ax^n + Bx^{-n}$ where A and B are arbitrary constants.

(ii) The Wronskian is given by

working+1
solution+1

$$\begin{aligned} \mathrm{Wr}\left[x^n, x^{-n}\right] &= x^n \frac{\mathrm{d}}{\mathrm{d}x}\left(x^{-n}\right) - \frac{\mathrm{d}}{\mathrm{d}x}\left(x^n\right)x^{-n} \\ &= -nx^nx^{-n-1} - nx^{n-1}x^{-n} \\ &= -nx^{-1} \end{aligned}$$

which is nonzero for $n \neq 0$ and hence the two solutions are linearly independent.

(b) For $n = 0$ the ODE reduces to $x^2y''(x) + xy'(x) = 0$. From part (a), we know that $y_1(x) = 1$ (i.e., x^0) is a solution. For reduction of order we trial a solution $y(x) = y_1(x)v(x) = v(x)$. Substituting into the ODE gives

trial sol+1

$$v''(x) + \frac{1}{x}v'(x) = 0$$

Making the substitution $w(x) = v'(x)$ gives

1st order ODE+1

$$w'(x) + \frac{1}{x}w(x) = 0$$

This can be solved with an integrating factor $\mathrm{IF}(x) = \mathrm{e}^{\int \frac{1}{x}\mathrm{d}x} = x$ and so

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}\left[xw(x)\right] &= 0 \\ xw(x) &= B \\ v'(x) &= \frac{B}{x} \\ v(x) &= A + B\log|x| \end{aligned}$$

Because our first solution was just a constant, the general solution to the ODE when $n = 0$ is therefore

$$y(x) = A + B\log|x|.$$

working+1
solution+1

Question 2.

cos bound+1
ratio (or other) test working+1

Solution.

(a) First we recognise that $|\cos(1 + n\pi x)| \leqslant 1$ for all n and x. Each term in the series is therefore bounded by $a_n = \dfrac{2^{-n}}{e^n}$. Now

ratio test result+1
W test +1

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{2^{-(n+1)}e^n}{2^{-n}e^{n+1}} = \frac{1}{2e^1} < 1$$

and so $\sum_{n=0}^{\infty} a_n$ converges by the ratio test and the series converges uniformly by the Weierstrass test.

(b) (i) Substituting the power series into the ODE gives

$$\sum_{n=0}^{\infty} \left[(1-x^2)c_n n(n-1)x^{n-2} - xc_n nx^{n-1} + c_n x^n\right] = 0$$

$$\sum_{n=0}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=0}^{\infty} c_n \left[-n(n-1) - n + 1\right] x^n = 0$$

$$\sum_{m=-2}^{\infty} c_{m+2}(m+1)(m+2)x^m + \sum_{n=0}^{\infty} c_n(1-n^2)x^n = 0 \qquad \text{(note the } m=-2 \text{ and } m=-1 \text{ terms are zero)}$$

$$\sum_{n=0}^{\infty} \left[c_{n+2}(n+1)(n+2) + c_n(1-n^2)\right] = 0.$$

correct substitution+1
working+1
correct recurrence +1
n=0/C2/C0 +1
n=1/C3 +1
gen soln+1

The recurrence relation is therefore $c_{n+2} = \dfrac{1-n^2}{(n+1)(n+2)}c_n = -\dfrac{n-1}{n+2}c_n$.

Now the $n = 0$ term in the sum gives

$$2c_2 + c_0 = 0 \qquad \Rightarrow \qquad c_2 = -\frac{1}{2}c_0 \qquad \text{where } c_0 \text{ is arbitrary.}$$

The $n = 1$ term in the sum gives

$$6c_3 = 0$$

which immediately sets all $c_{2m-1} = 0$ for $m = 2, 3, 4, \ldots$ but leaves c_1 arbitrary. Therefore x is a solution to the ODE. By making the substitution $n = 2m$, the general solution is then

$$y(x) = Ax + \sum_{m=0}^{\infty} c_m x^{2m}, \qquad \text{where} \qquad c_{m+1} = -\frac{2m-1}{2m+2}c_m \qquad \text{with arbitrary } c_0$$

(ii) We can find the interval of convergence for the series solution using the ratio test

$$\lim_{n\to\infty} \left|\frac{c_{m+1}x^{2(m+1)}}{c_m x^{2m}}\right| < 1$$

$$\lim_{n\to\infty} \left|\frac{2m-1}{2m+2}x^2\right| < 1$$

$$\lim_{m\to\infty} \left|\frac{2-1/m}{2+2/m}\right| x^2 < 1$$

$$x^2 < 1$$

$$|x| < 1$$

ratio test working+1
ratio test result+1
Interval +1

i.e., the interval of convergence is -1 to 1.

(iii) Re arranging the ODE to standard form we have

$$y''(x) - \frac{x}{1-x^2}y'(x) + \frac{1}{1-x^2}y(x) = 0$$

To guarantee a unique solution, the coefficient functions must be continuous on an interval containing the initial condition. Because the initial condition is at $x = 0.2$, we are guaranteed a unique solution on the interval $x \in (-1, 1)$. Aside: given this is also the interval of convergence of the series solution in (ii), applying the initial condition to that solution will give the unique solution to this IVP.

continuity of coeff functions +1
Initial condition & interval +1

Question 3.

Vector-matrix system +1

Solution.

(a) The vector valued system is

$$\boldsymbol{u}'(t) + \mathrm{A}\boldsymbol{u}(t) = \boldsymbol{b} \qquad \text{where} \qquad \mathrm{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{u}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(b) First we find the eigensystem for A. The eigenvalues are

$$(2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0$$

and so $\lambda_1 = 3$ and $\lambda_2 = 1$. To find the eigenvectors:

$$(\mathrm{A} - 3\mathrm{I})\boldsymbol{v}_1 = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} = 0 \qquad\qquad (\mathrm{A} - \mathrm{I})\boldsymbol{v}_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 0$$

and so possible eigenvectors are $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. The matrix can therefore be diagonalised by

$$\mathrm{A} = \mathrm{PDP}^{-1} \qquad \text{where} \qquad \mathrm{D} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \mathrm{P} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \qquad \text{and} \qquad \mathrm{P}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Multiplying the vector equation by P^{-1} gives

$$\mathrm{P}^{-1}\boldsymbol{u}'(t) + \mathrm{P}^{-1}\mathrm{PDP}^{-1}\boldsymbol{u}(t) = \mathrm{P}^{-1}\boldsymbol{b}$$

We now let $\boldsymbol{z}(t) = \mathrm{P}_{-1}\boldsymbol{u}(t)$ so that

$$\boldsymbol{z}'(t) + \mathrm{D}\boldsymbol{z}(t) = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{z}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We therefore have two decoupled ODEs to solve, $z_1(t) + 3z_1(t) = 2$ with $z_1(0) = 0$ and $z_2(t) + z_2(t) = 1$ with $z_2(0) = 0$. We can solve using an integrating factor

$$\mathrm{IF}_1(t) = \mathrm{e}^{\int 3\mathrm{d}t} = \mathrm{e}^{3t} \qquad \text{and} \qquad \mathrm{IF}_2(t) = \mathrm{e}^{t}$$

so that we have the equations

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}[\mathrm{e}^{3t}z_1(t)] &= 2\mathrm{e}^{3t} & \text{and} \qquad \frac{\mathrm{d}}{\mathrm{d}t}[\mathrm{e}^{t}z_1(t)] &= \mathrm{e}^{t} \\ \mathrm{e}^{3t}z_1(t) &= \frac{2}{3}\mathrm{e}^{3t} + A & \text{and} \qquad \mathrm{e}^{t}z_2(t) &= \mathrm{e}^{t} + B \\ z_1(t) &= \frac{2}{3} + A\mathrm{e}^{3t} & \text{and} \qquad z_2(t) &= 1 + B\mathrm{e}^{-t} \end{aligned}$$

The initial condition implies $A = -\frac{2}{3}$ and $B = -1$ and so $z_1(t) = \frac{2}{3}\left(1 - \mathrm{e}^{-3t}\right)$ and $z_2(t) = 1 - \mathrm{e}^{-t}$. Now since $\boldsymbol{u}(t) = \mathrm{P}\boldsymbol{z}(t)$, we have

$$\boldsymbol{u}(t) = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} \frac{2}{3}\left(1 - \mathrm{e}^{-3t}\right) \\ 1 - \mathrm{e}^{-t} \end{bmatrix}$$

evalues working +1
correct evalues +1
e-vector +2
matrix D +1
Matrix P&Inverse +1
1st system in Z + 1
Z1 + 2
Z2 +2
ICs +2
Final solution +1

Question 4.

Solution.

(a) (i) We have

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{s+2}{s^2+2s+10}\right\}(s) &= \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+9}\right\}(s) + \mathcal{L}_{-1}\left\{\frac{1}{(s+1)^2+9}\right\}(s)\\ &= \mathrm{e}^{-t}\left(\mathcal{L}^{-1}\frac{s}{s^2+3^2} + \frac{1}{3}\mathcal{L}^{-1}\frac{3}{s^2+3^2}\right)\\ &= \mathrm{e}^{-t}\left(\cos 3t + \frac{1}{3}\sin 3t\right)\end{aligned}$$

split to two shifts +1
s-shifts +2
partial decomp +1
coefficients +3
Inversion +2

(ii) We must use partial fractions here

$$\begin{aligned}\frac{1}{s(s^2+2s+10)} &= \frac{a}{s} + \frac{bs+c}{s^2+2s+10}\\ \Rightarrow \qquad 1 &= as^2 + 2as + 10a + bs^2 + cs\end{aligned}$$

We immediately have $a = \frac{1}{10}$, $a + b = 0$ therefore $b = -\frac{1}{10}$ and $2a + c = 0$ therefore $c = -\frac{1}{5}$. Now we have

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+2s+10)}\right\} &= \mathcal{L}^{-1}\left\{\frac{1}{10}\frac{1}{s} - \frac{1}{10}\frac{s+2}{s^2+2s+10}\right\}\\ &= \frac{1}{10} - \frac{1}{10}\mathrm{e}^{-t}\left(\cos 3t + \frac{1}{3}\sin 3t\right)\end{aligned}$$

(b) (i) The forcing term is a pulse that switches on for five units of time before being switched off. The initial conditions state the system starts from its unforced equilibrium position with zero velocity.

(ii) Taking the Laplace transform of both sides gives

$$\begin{aligned}s^2\mathcal{L}y - sy(0) - y'(0) + 2[s\mathcal{L}y - y(0)] + 10\mathcal{L}y &= \frac{1-\mathrm{e}^{-5t}}{s}\\ (s^2+2s+10)\mathcal{L}y &= \frac{1-\mathrm{e}^{-5s}}{s}\\ \mathcal{L}y &= \frac{1-\mathrm{e}^{-5s}}{s(s^2+2s+10)}\end{aligned}$$

Upon inversion this becomes

$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s(s^2+2s+10)}\right\} - \mathcal{L}^{-1}\left\{\frac{\mathrm{e}^{-5s}}{s(s^2+2s+10)}\right\}$$

on/off forcing term +1
ICs +1

$$y(t) = \frac{1}{10} - \frac{1}{10}\mathrm{e}^{-t}\left(\cos 3t + \frac{1}{3}\sin 3t\right) + u(t-5)\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+2s+10)}\right\}(t-5)$$

LT of ODE +1
working +1
Correct Y(s) +1
Inversion +2

$$y(t) = \frac{1}{10} - \frac{1}{10}\mathrm{e}^{-t}\left(\cos 3t + \frac{1}{3}\sin 3t\right) + u(t-5)\left[\frac{1}{10} - \frac{1}{10}\mathrm{e}^{-(t-5)}\left(\cos 3(t-5) + \frac{1}{3}\sin 3(t-5)\right)\right]$$

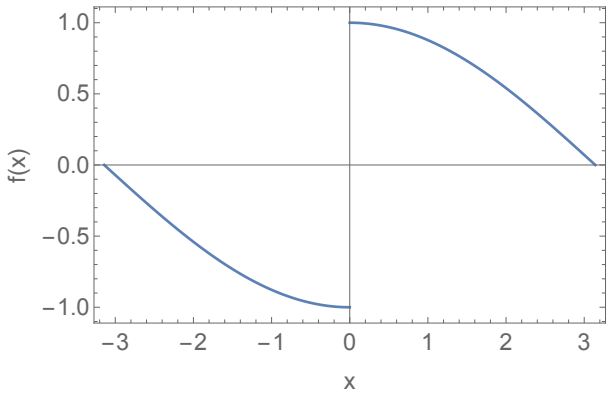
$$y(t) = \begin{cases}\frac{1}{10} - \frac{1}{10}\mathrm{e}^{-t}\left(\cos 3t + \frac{1}{3}\sin 3t\right) & t \leqslant 5\\ \frac{1}{5} - \frac{1}{10}\mathrm{e}^{-t}\left(\cos 3t + \frac{1}{3}\sin 3t\right) - \frac{1}{10}\mathrm{e}^{-(t-5)}\left(\cos 3(t-5) + \frac{1}{3}\sin 3(t-5)\right) & t > 5\end{cases}$$

t-shift of 2nd term +1
Final answer without unit step + 1

Question 5.

Solution.

(a)



shape +1
axes labels + 1

(b) $f(x)$ is odd

answer + 1

(c) Because $f(x)$ is odd with $L = \pi$, the Fourier series representation of $f(x)$ is given by $g(x) = \sum_{n=1}^{\infty} b_n \sin nx$ where

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi} \cos\left(\frac{x}{2}\right)\sin(nx)\mathrm{d}x \\ &= \frac{2}{\pi}\int_0^{\pi} \frac{1}{2}\left[\sin\left(nx - \frac{x}{2}\right) + \sin\left(nx + \frac{x}{2}\right)\right]\mathrm{d}x \\ &= \frac{1}{\pi}\int_0^{\pi} \sin\left(n - \frac{1}{2}\right)x + \sin\left(n + \frac{1}{2}\right)x\mathrm{d}x \\ &= -\frac{1}{\pi}\left[\frac{1}{n-\frac{1}{2}}\cos\left(n - \frac{1}{2}\right)x + \frac{1}{n+\frac{1}{2}}\cos\left(n + \frac{1}{2}\right)x\right]_0^{\pi} \end{aligned}$$

correct integral +1
working + 1
correct coeff bn +1

but $\cos\left(n - \frac{1}{2}\right)\pi = 0$ and $\cos\left(n + \frac{1}{2}\right)\pi = 0$ so

$$b_n = \frac{1}{\pi}\left(\frac{1}{n-\frac{1}{2}} + \frac{1}{n+\frac{1}{2}}\right)$$

No further simplification from here is necessary but the result can be simplified to $b_n = \frac{8}{\pi}\frac{n}{4n^2-1}$

(d) (i) The Fourier coefficients decay as $\frac{1}{n}$ for large n.

(ii) $g(x)$ will NOT converge uniformly since the periodic extension of $f(x)$ is discontinuous

(iii) $g(x)$ will exhibit Gibbs phenomena at $x = 0$. The gap width here is 2 and, as the error is approximately 9% of the gap width, the error will be approximately 1.8. (The values of 10%, 8.9% or anything else close to these values is also acceptable).

(e) The periodic extension of $h(x)$ is continuous, therefore the Fourier series of $h(x)$ will converge uniformly.

(i) answer +1
(ii) answer/explanation + 2
(iii) x=0 and error +2
(e) answer/explanation + 2

Question 6.

Solution.

(a) (i) From the definition of the Fourier transform

$$\widehat{f}(k) = \int_{-\infty}^{\infty} \delta_2(x) e^{-ikx} dx = e^{-2ik}$$

(ii) Similarly

$$\widehat{g}(k) = \int_{-\infty}^{\infty} \delta_{-2}(x) e^{-ikx} dx = e^{2ik}$$

(b) We can rewrite cos in terms of complex exponentials: $\cos 2k = \frac{1}{2}\left(e^{-2ik} + e^{2ik}\right)$. The inverse Fourier transform of $\cos 2k$ is therefore

$$h(x) = \frac{1}{2}\left(\delta_2(x) + \delta_{-2}(x)\right)$$

(i) answer + 1
(ii) answer + 1
(b) working/answer + 2
(c) working + 2
IVP + 1
(d) state both steps +1

(c) Transforming the PDE into Fourier space gives

$$\frac{d^2}{dt^2}\widehat{u}(k,t) = -k^2c^2\widehat{u}(k,t).$$

We must also transform the two initial conditions:

$$\widehat{u}(k,0) = \mathscr{F}\{\delta_2(x) + \delta_{-2}(x)\} = e^{-2ik} + e^{2ik} = 2\cos 2k$$
$$\left.\frac{d\widehat{u}}{dt}\right|_{t=0} = 0.$$

This fully specifies an IVP that we may then solve for $\widehat{u}(k,t)$.

(d) The remaining two steps to solve the problem are (i), solve the IVP for $\widehat{u}$ and (ii), invert the Fourier transform to find $u(x,t)$.

Question 7.

Solution.

(a) (i) First we let $u(r,\theta) = R(r)S(\theta)$. We then substitute into the steady state heat equation, i.e., where $\frac{\partial u}{\partial t} = 0$,

(a, i)
soln form +1
substitution & working +1
constant and ODEs +1

+1 correct PDE

$$R''(r)S(\theta) + \frac{1}{r}R'(r)S(\theta) + \frac{1}{r^2}R(r)S''(\theta) = 0$$
$$r^2\frac{R''(r)}{R(r)} + r\frac{R'(r)}{R(r)} = -\frac{S''(\theta)}{S(\theta)}$$

Now since LHS depends only on r and RHS only on θ, this can only hold if both sides are equal to the same constant, i.e.,

+1 (one for each ODE)

$$r^2\frac{R''(r)}{R(r)} + r\frac{R'(r)}{R(r)} = \lambda \qquad -\frac{S''(\theta)}{S(\theta)} = \lambda$$
$$r^2R''(r) + rR'(r) - \lambda R(r) = 0 \qquad S''(\theta) + \lambda S(\theta) = 0$$

(ii) Assuming that $\lambda > 0$, we set $\lambda = k^2$ which gives the solutions $S(\theta) = A\cos k\theta + B\sin k\theta$. The Dirichlet boundary conditions lead to the eigenfunctions:

$$S_n(\theta) = \sin n\theta \qquad \text{where} \qquad n = 1, 2, 3, \ldots$$

Solution to S +3

The $R(r)$ equation and solution is therefore

$$r^2R_n''(r) + rR'(r) - n^2R(r) = 0$$
$$R_n(r) = A_nr^n + B_nr^{-n}$$

Solution to R +1
+1

where A_n and B_n are constants. The general solution is therefore

$$u(r,\theta) = \sum_{n=1}^{\infty}\left(A_nr^n + B_nr^{-n}\right)\sin n\theta.$$

gen soln +1
BC for r=1 +1
+1

Application of the $r = 1$ boundary condition gives

$$u(1,\theta) = \sum_{n=1}^{\infty}(A_n + B_n)\sin n\theta = 0$$

and so from orthogonality we know that $A_n + B_n = 0$ for all n. We therefore have

$$u(r,\theta) = \sum_{n=1}^{\infty}A_n(r^n - r^{-n})\sin n\theta.$$

Application of the $r = 2$ boundary condition gives

$$u(2,\theta) = \sum_{n=1}^{\infty}A_n(2^n - 2^{-n})\sin n\theta = \frac{\pi}{2}\sin 4\theta - \sin 7\theta.$$

By orthogonality we have that $A_4\left(2^4 - 2^{-4}\right) = \frac{\pi}{2}$ and $A_7\left(2^7 - 2^{-7}\right) = 1$ so the final solution is therefore

$$u(r,\theta) = \frac{\pi}{2}\frac{r^4 - r^{-4}}{2^4 - 2^{-4}}\sin 4\theta + \frac{r^7 - r^{-7}}{2^7 - 2^{-7}}\sin 7\theta.$$

BC for r=2 +1
working for 2 constants +2
gen soln +1

(b) (i) The total heat is

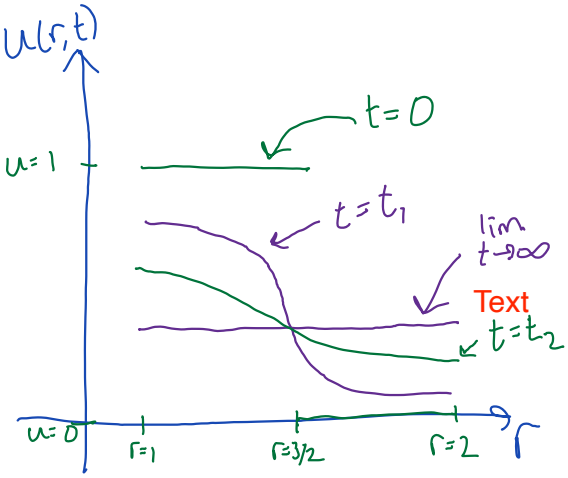
Note this is the answer to an easier question than was featured in the exam but gives you an idea of the mark allocation and working. You are left to check your work on this part using this solution as a guide.

Here u(r.0)=1 for r in [1,1.5] and zero elsewhere.

$$\int_1^2\int_0^{\pi}u(r,0)r\mathrm{d}\theta\mathrm{d}r = \pi\int_1^{3/2}r\mathrm{d}r = \pi\left[\frac{r^2}{2}\right]_1^{3/2} = \frac{\pi}{2}\left(\left(\frac{3}{2}\right)^2 - 1\right) = \frac{5\pi}{8}$$

+1 for Jacobian
correct integral +1
working +1
answer + 1

(ii)



shape +2
labels +1
+1

(c) The series' $\sum_{n=0}^{\infty}u_n(\theta,t)$, $\sum_{n=0}^{\infty}\frac{\partial u_n}{\partial t}$, $\sum_{n=0}^{\infty}\frac{\partial u_n}{\partial \theta}$ and $\sum_{n=0}^{\infty}\frac{\partial^2 u_n}{\partial \theta^2}$ must all converge uniformly if $u(\theta,t)$ is a genuine solution.

List all series +1
+2